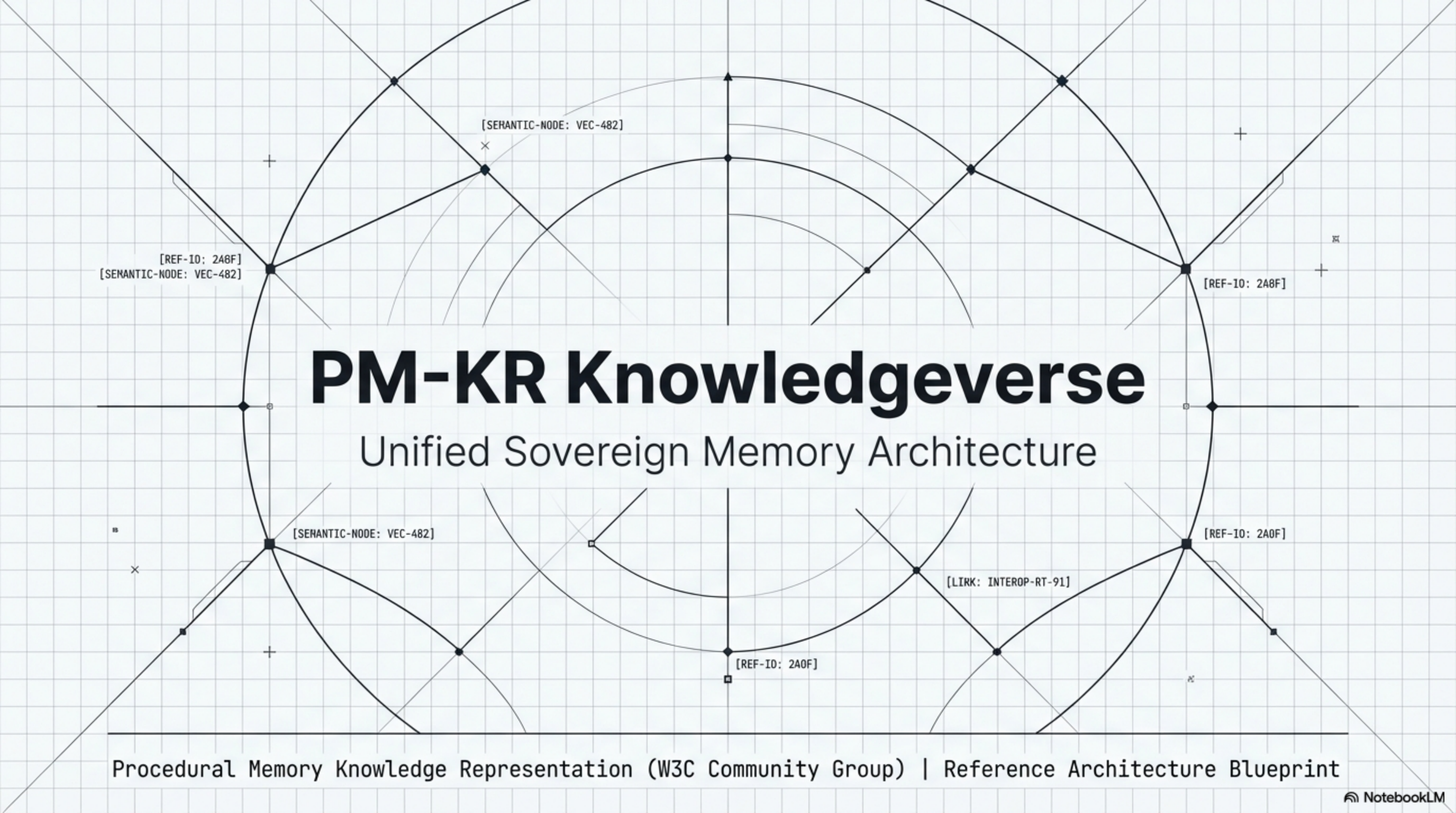


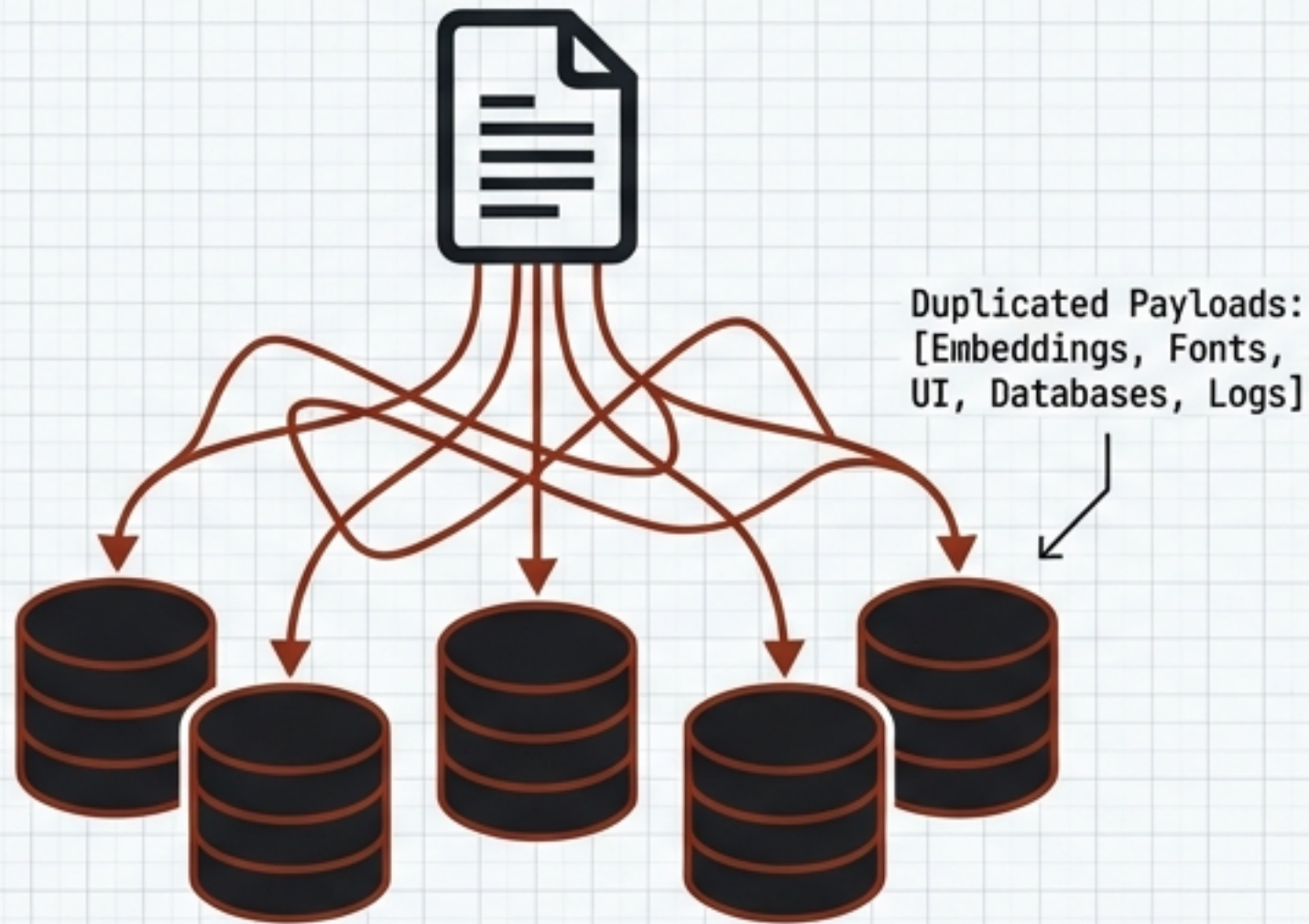


PM-KR Knowledgeverse

Unified Sovereign Memory Architecture

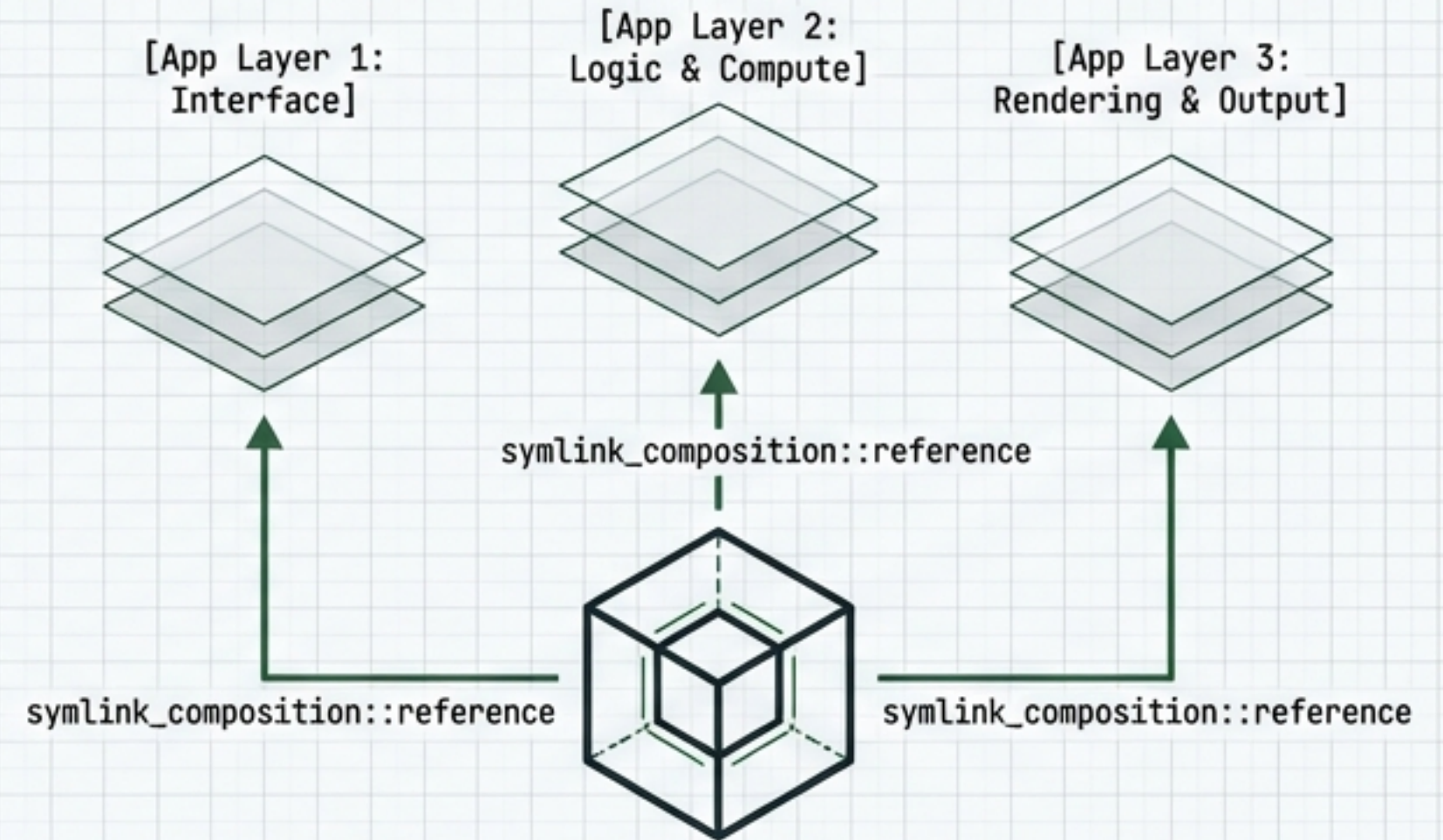
The Core Paradigm Shift: From Static Payloads to Procedural Executables

THE PAST



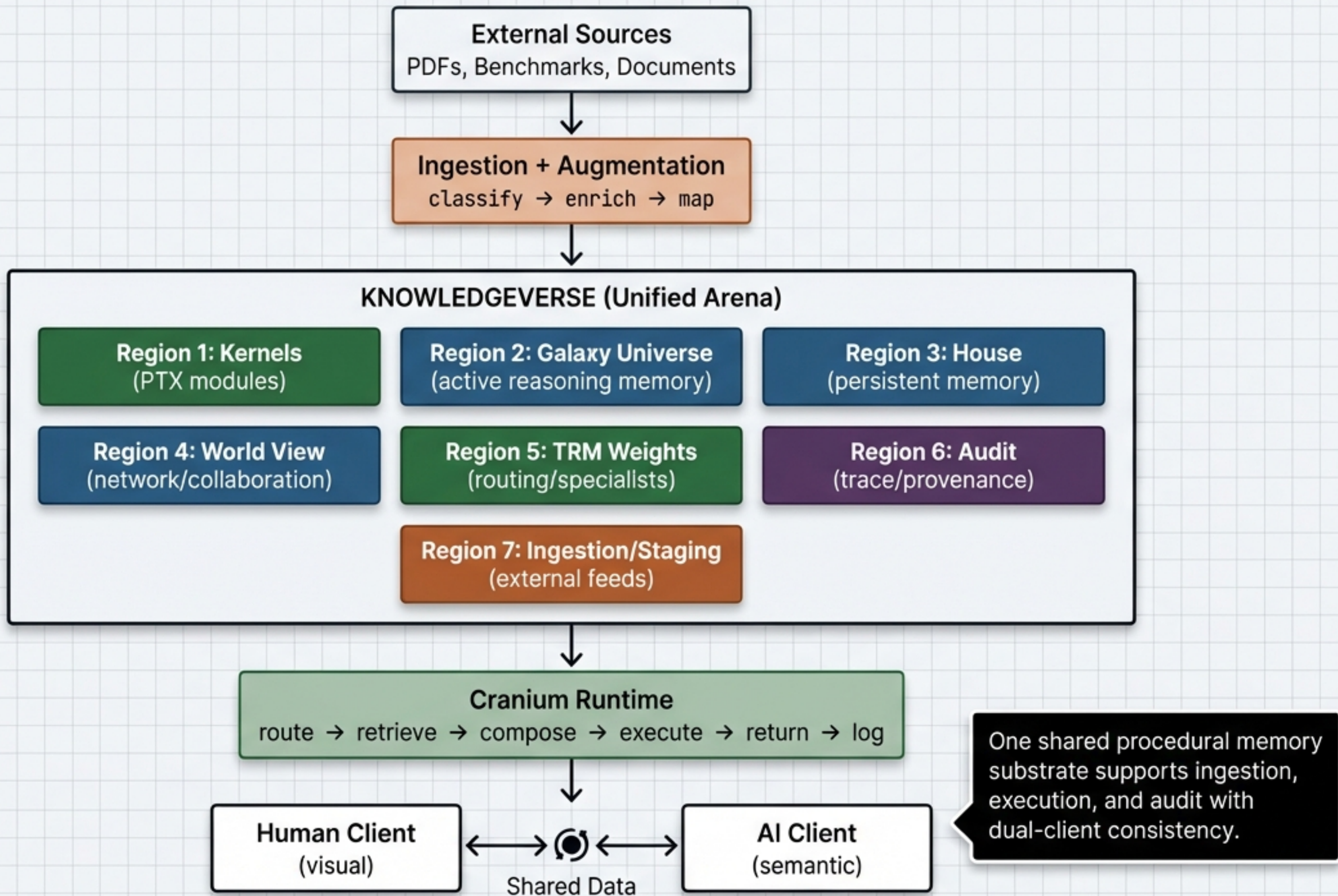
[The Problem] Current AI copies data everywhere (drifting, expensive, opaque).

PM-KR

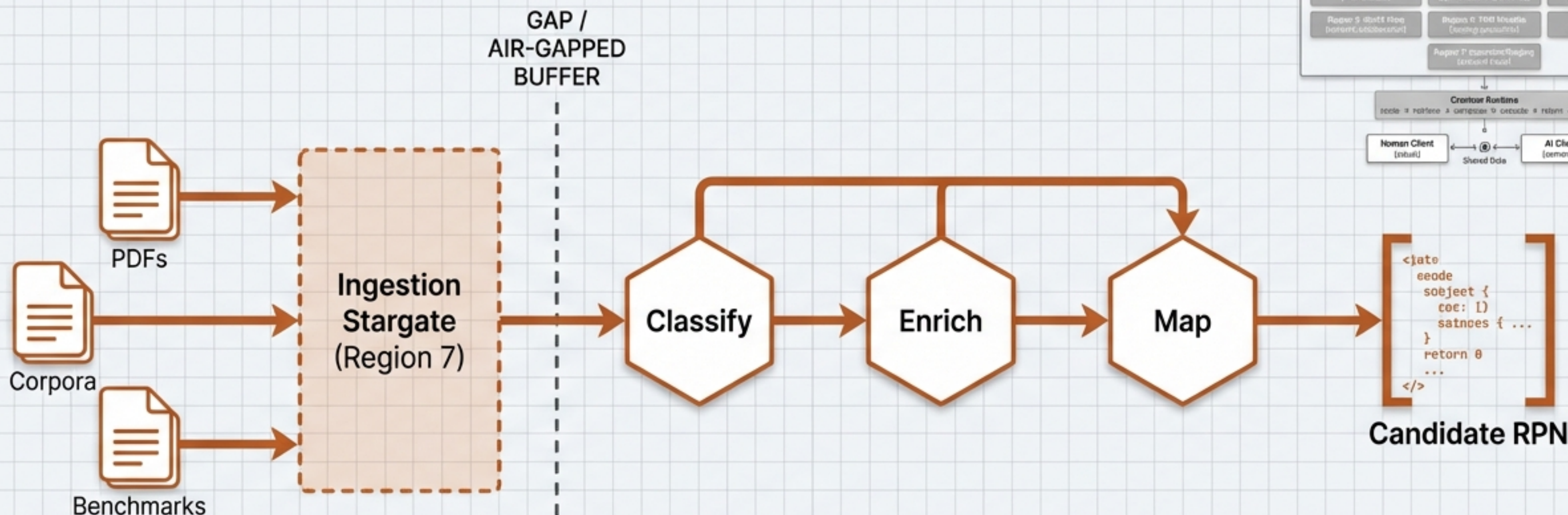


[The PM-KR Solution] “Save Information Principle” — Store knowledge once as executable procedures (RPN). Reference it via symlink-style composition.

[Result] Zero duplication, deterministic traces, and sovereign hot-path execution.



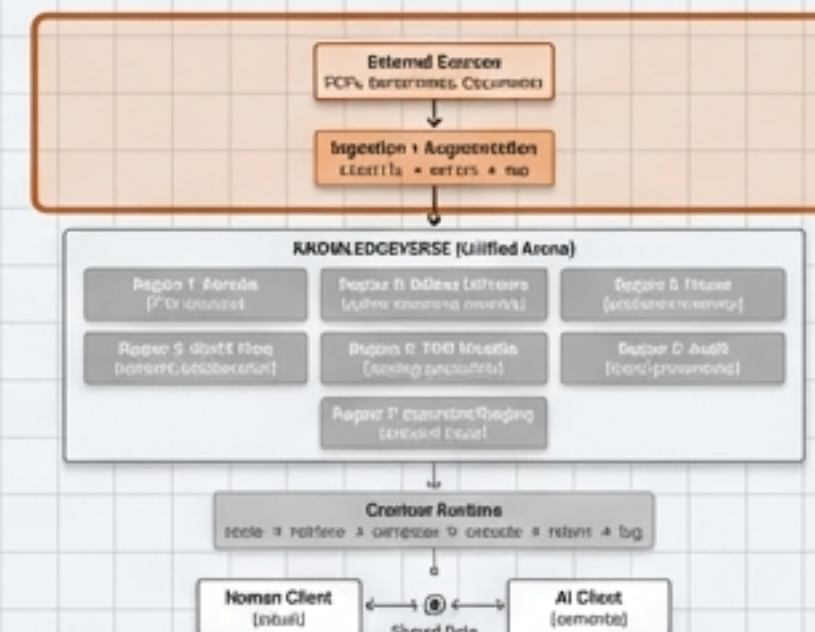
Layer 1 & 2: External Sources to Ingestion Stargate



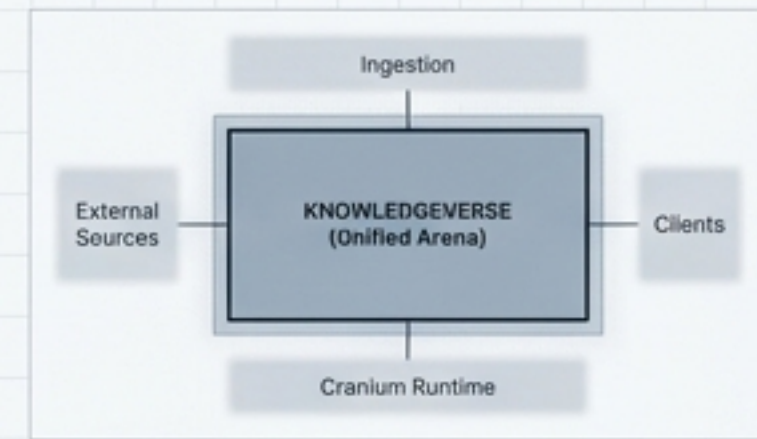
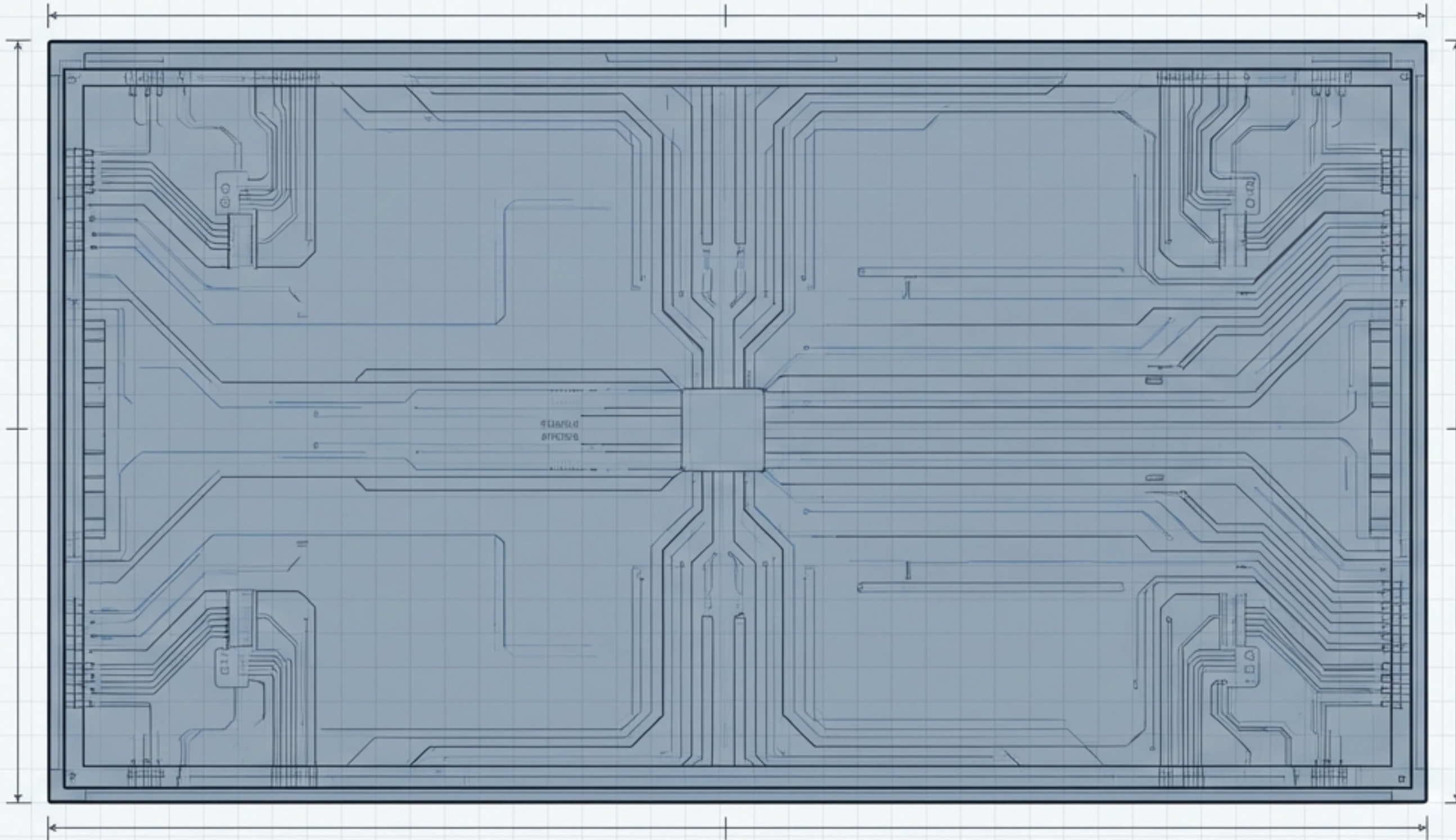
[Process]: PDFs, corpora, and benchmarks hit the Region 7 Stargate buffer.

[Transformation]: Raw data → Classify
→ Enrich → Map → Candidate RPN
(Reverse Polish Notation) programs.

[Integrity]: Resumable page staging, skip-source logs for encrypted files, deterministic rebuilds from stage state.



Layer 3: The Knowledgeverse Substrate



[One Persistent PTX Context]

8.5GB VRAM (on a 12GB baseline) dedicated as a single unified arena.

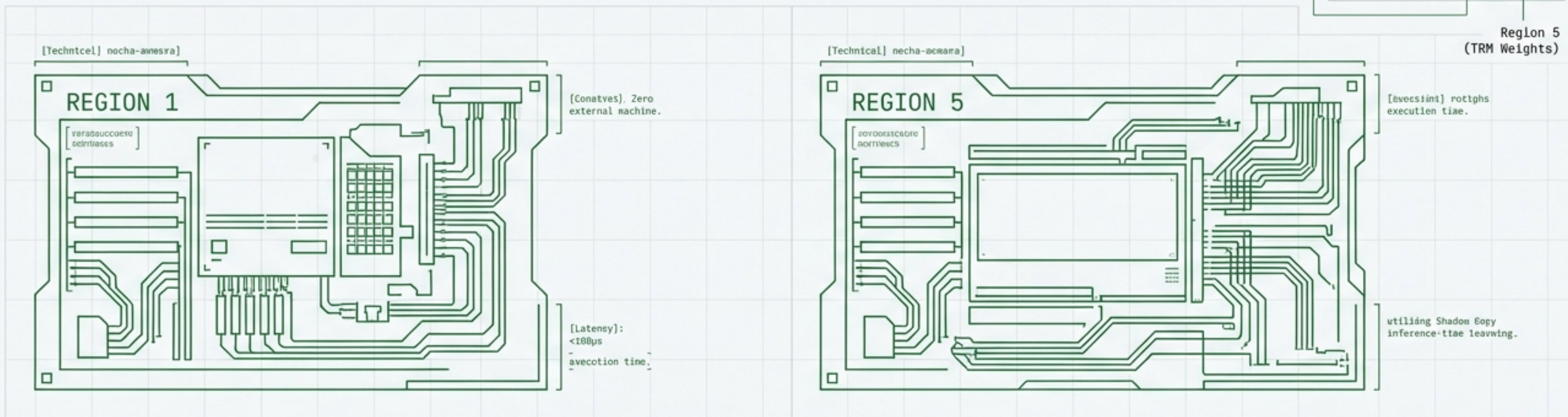
[Why?]

Eliminates CUDA context switching conflicts between PyTorch, sovereign loaders, and external libraries.

[Architecture]

Deterministic boot sequence with hash-bound artifacts. Memory fragmentation is eliminated via strict region allocation (PINNED, WARM, COLD, REBUILDABLE).

Region 1 & 5: Sovereign Execution Core



Region 1: Kernels (PINNED)

- 42+ hand-written PTX modules.

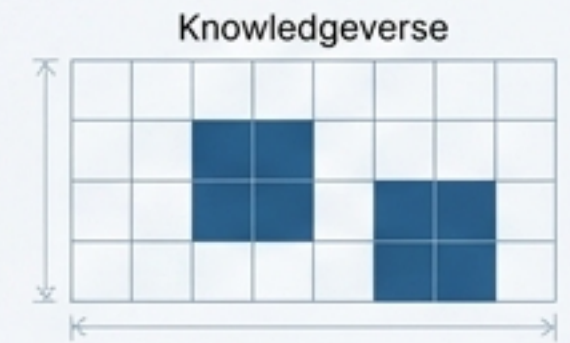
[Constraint]: Zero external machine learning framework dependencies.

[Latency]: <100µs median execution time.

Region 5: TRM Weights

- Base Tiny Recursive Model (~7M parameters, 10,000x more efficient than 70B LLMs).
- Specialist routing adapters (math, visual, physics) utilizing Shadow Copy inference-time learning.

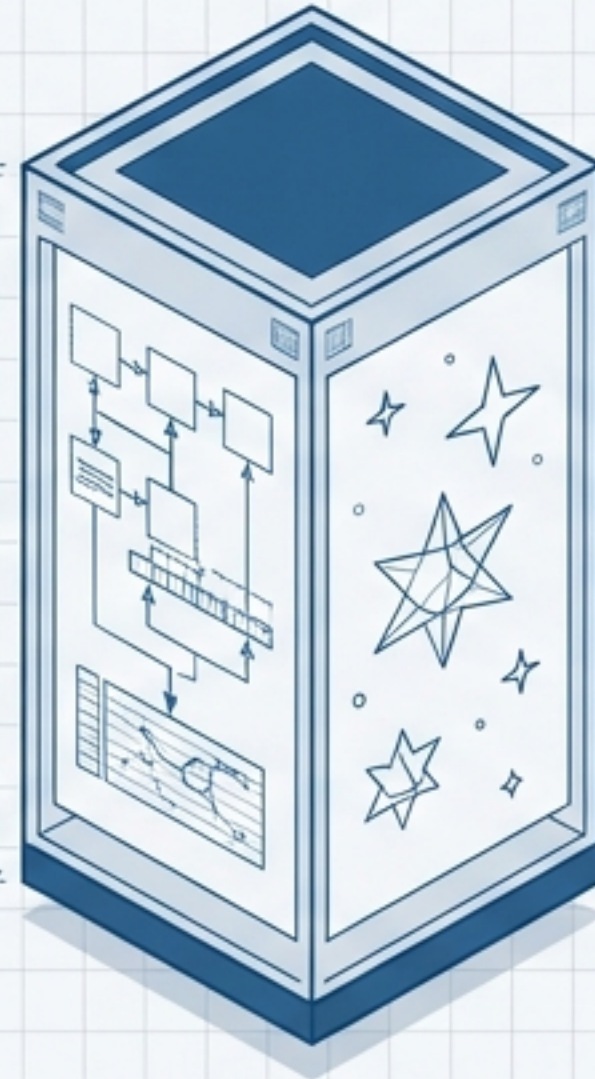
Region 2 & 3: The Dual-Memory Core



Region 2: Galaxy Universe (Active RAM)

- Volatile VRAM workspace.
- High-dimensional space where knowledge floats as geometric stars in a semantic starfield.

[Access]: $\sim 5\mu\text{s}$ latency.

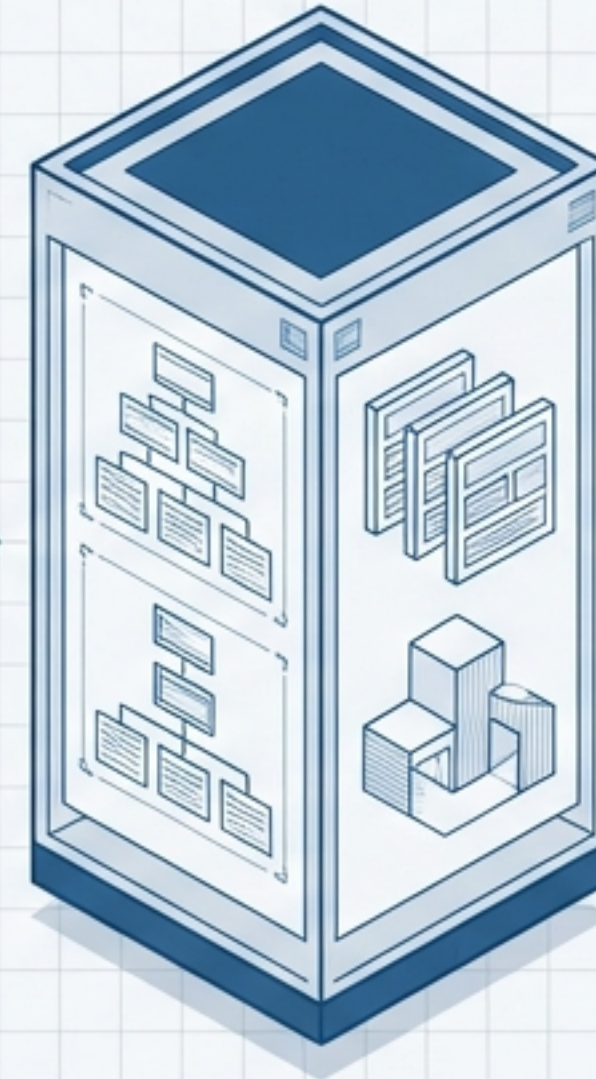


SleepTime
Consolidation

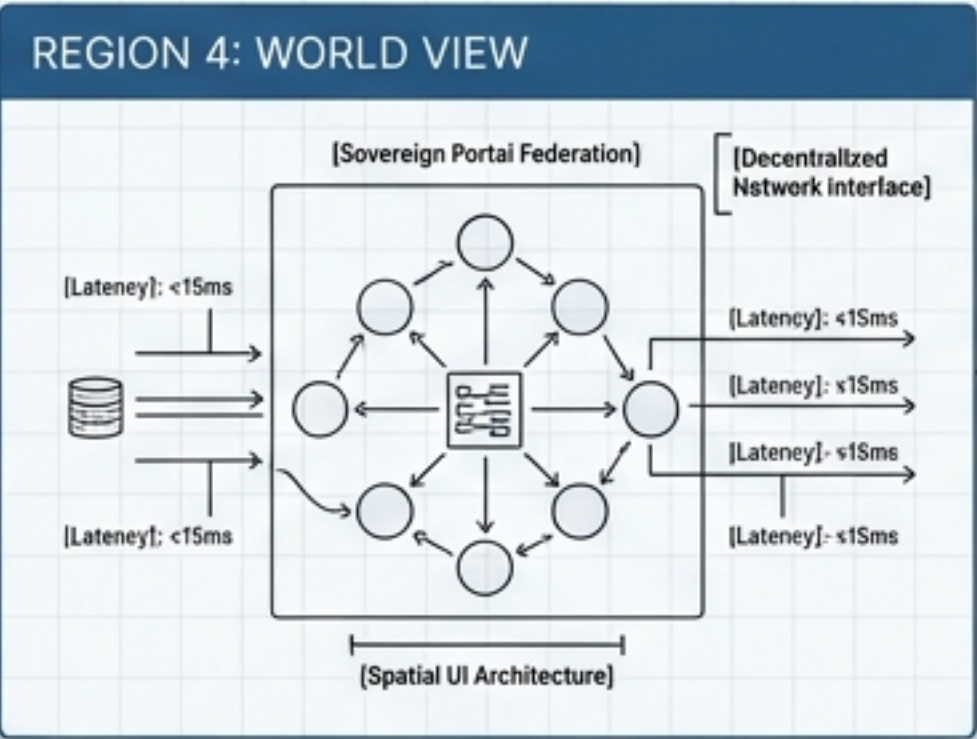
Region 3: House Context (Persistent Disk)

- Stable, low-dimensional gITF SSD storage.
- Consolidated knowledge embodied as 3D spatial structures (Virtual Libraries, Fractal Knowledge Gardens).

[Access]: $\sim 5\text{ms}$ latency.

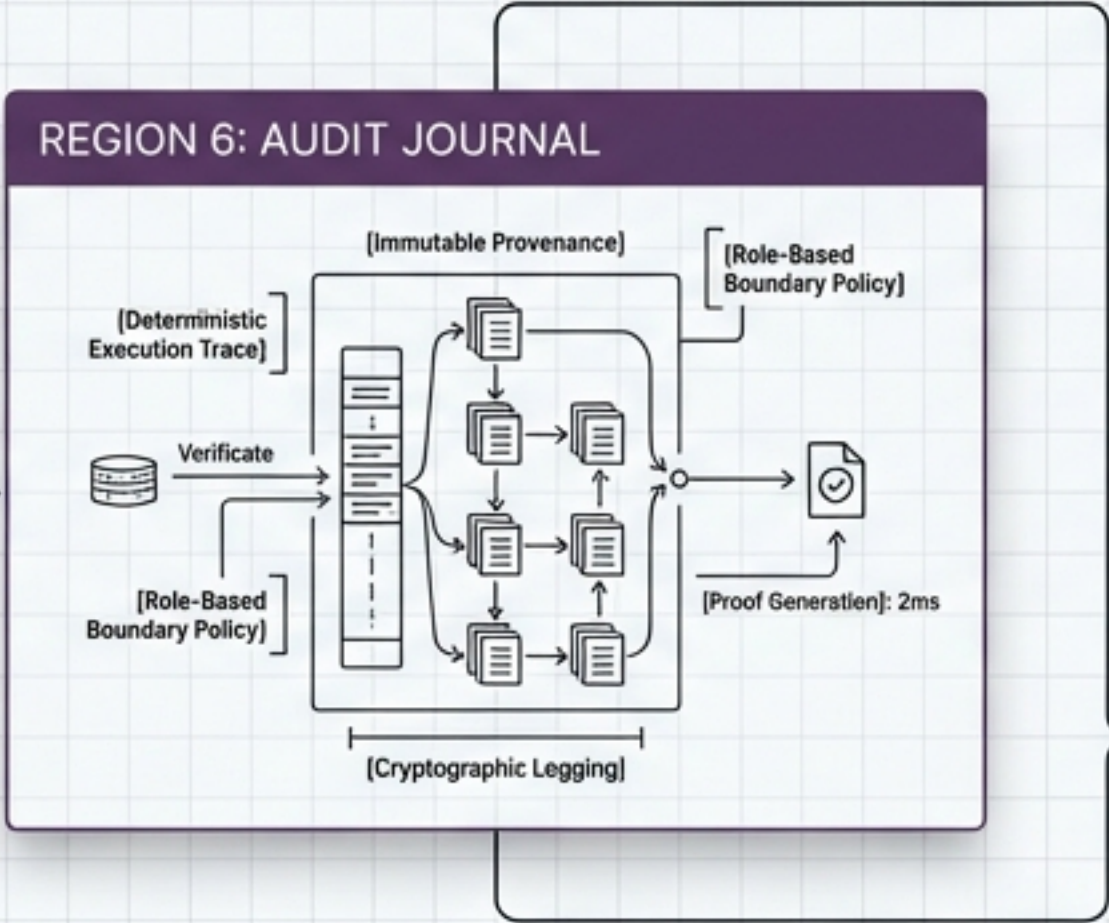


Region 4, 6, 7: Network, Audit & Staging

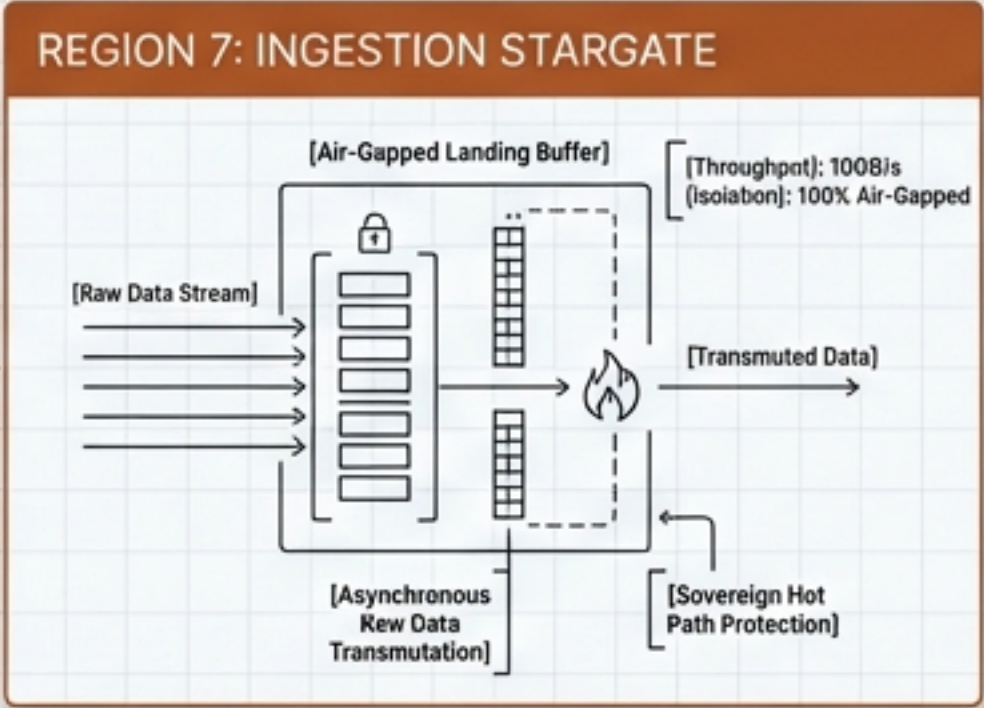


Region 4: World View — Spatial UI architecture, Portal Federation, and decentralized network streaming and cryptographic logging.

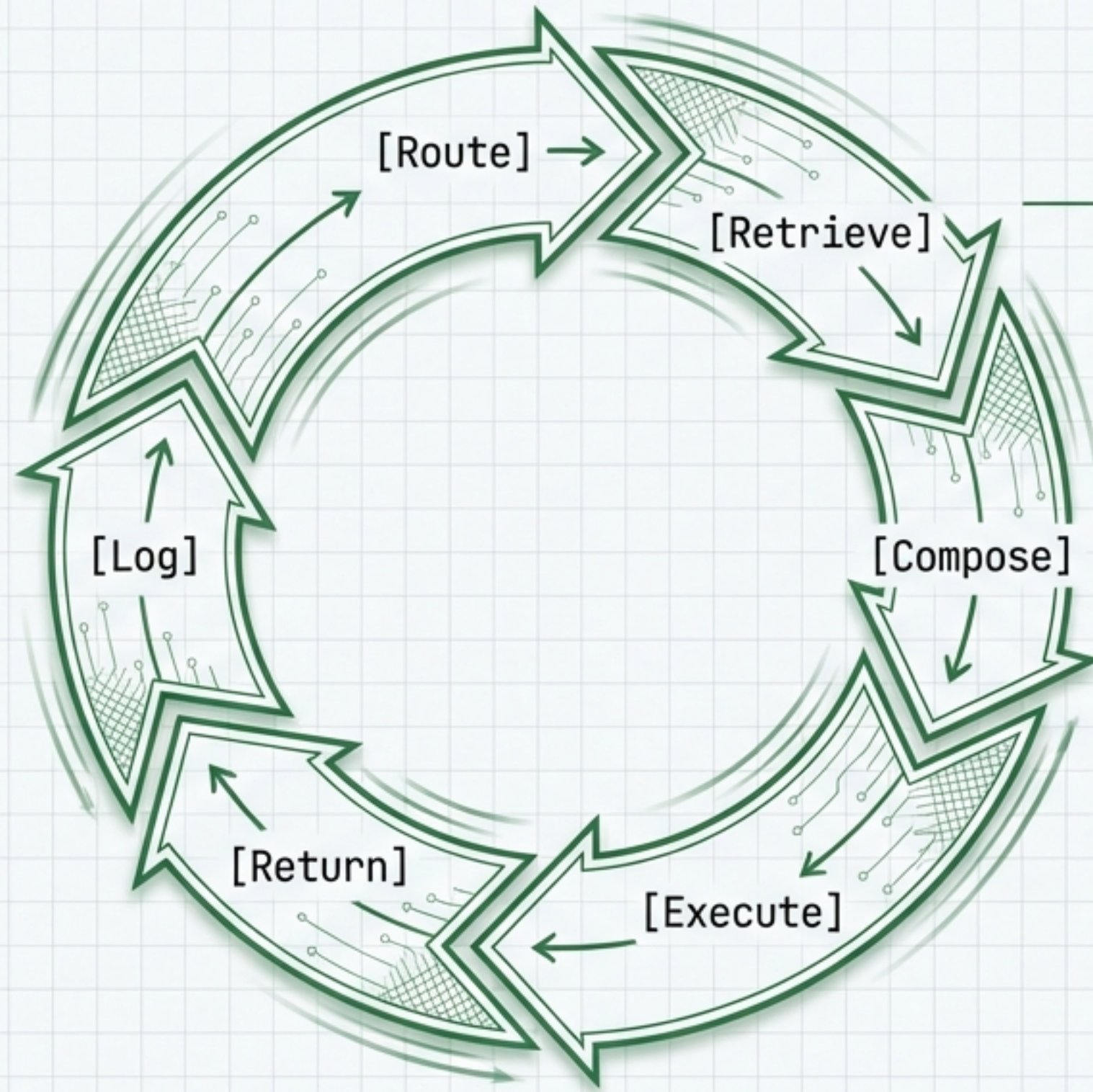
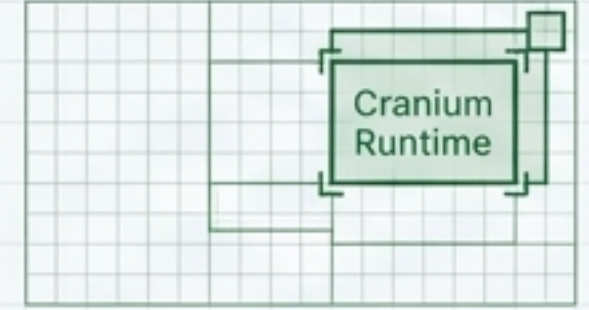
Region 4: World View — Spatial UI architecture, Portal Federation, and decentralized network streaming interfaces.



Region 7: Ingestion Stargate — Air-gapped landing buffer for asynchronous raw data transmutation, protecting the sovereign hot path.



Layer 4: Cranium Runtime Hot Path



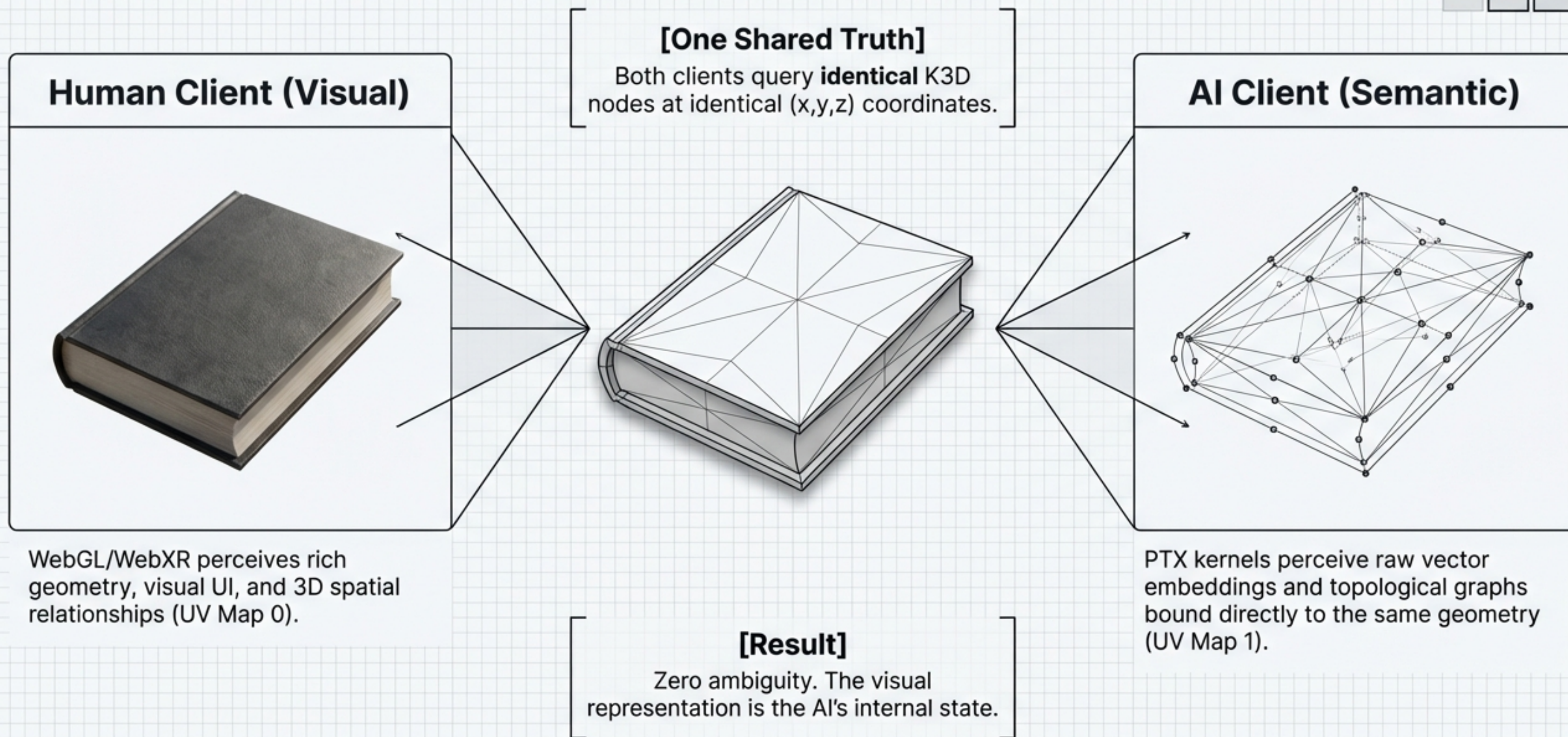
• [The Engine]

PTX-backed Reverse Polish Notation (RPN) VM stack (18 parallel instances, 69-instruction limits).

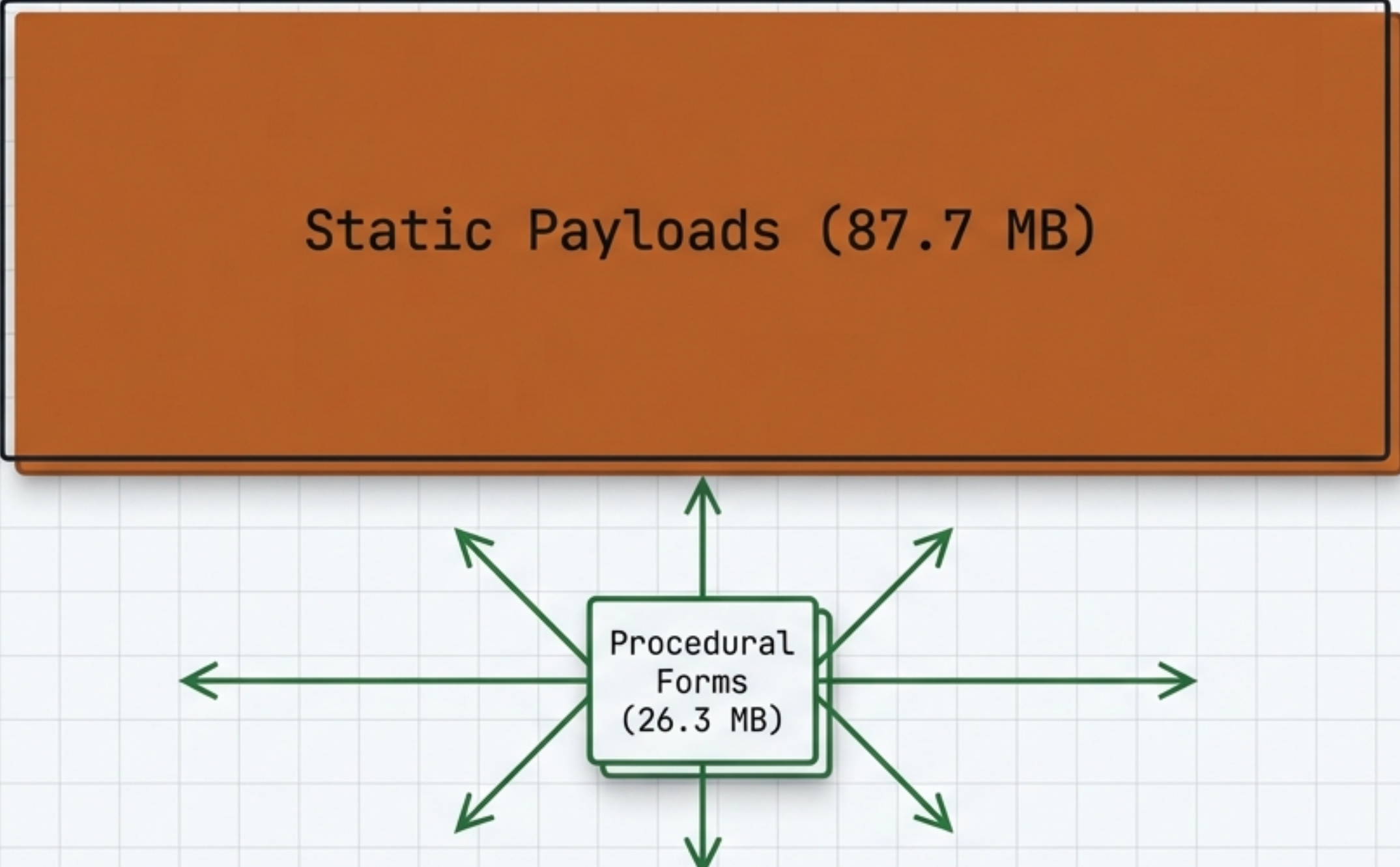
• [The Sovereign Invariant]

The hot path is 100% GPU-native. Zero CPU fallbacks. Zero external library dependencies. Intelligence is executed, not stored.

Layer 5: The Dual-Client Reality



The Output: 70%+ Procedural Compression



The diagram illustrates the concept of procedural compression. At the top, a large orange rectangle is labeled 'Static Payloads (87.7 MB)'. Below it, a smaller white rectangle with a green border is labeled 'Procedural Forms (26.3 MB)'. Eight green arrows radiate from the 'Procedural Forms' box, pointing towards the 'Static Payloads' box, indicating that the procedural forms are used to generate or reference the static payloads.

Static Payloads (87.7 MB)

[Symlink-Style Composition]

Instead of duplicating knowledge, the system stores one canonical procedural RPN program and references it infinitely.

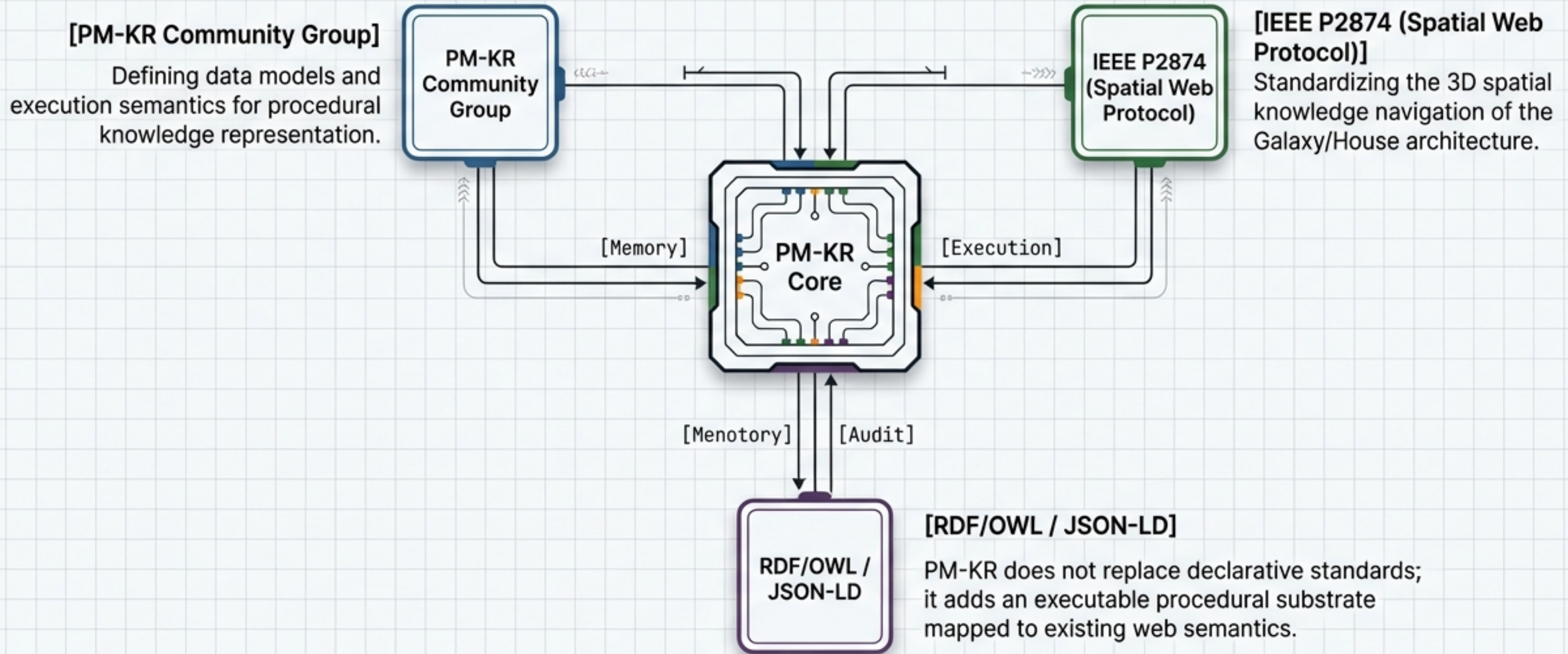
[Example]

Character Galaxy compressed from 87.7 MB (duplicated font payload) to 26.3 MB (procedural composition).

[Scale]

Capable of holding 51,532 complex nodes in <200MB of VRAM.

W3C Alignment & Standards Track



The PM-KR Knowledgeverse

